Foundations of Large Language Models

Abilities and Limitations through a Formal Language Learning Perspective

Bishwamittra Ghosh **Krishna P. Gummadi**

https://courses.mpi-sws.org/fllm-ws26/

Our Team



Bishwamittra Ghosh

Instructor



Krishna P. Gummadi

Instructor



Pushpdeep Singh

Teaching Assistant

Large Language Models

A large language model is a parameter-rich neural network trained to model the probability distribution of sequences of words, enabling it to generate, predict, and understand text across diverse contexts.

$$P(s_t \mid s_{t-1}, \ldots, s_1)$$

Large Language Models

A large language model is a parameter-rich neural network trained to model the probability distribution of sequences of words, enabling it to generate, predict, and understand text across diverse contexts.

$$P(s_t \mid s_{t-1}, \ldots, s_1)$$

Conditional Probability Table: A Faithful Simulation of an LLM

Prefix	Generation	Probability
Germany's capital is	Paris	0.02
	Berlin	0.91
	Washington DC	0.01
	...	...

Vocabulary: $50,000$ tokens
Context length: 2048 tokens
Unique entries $50,000^{2048} \gg 10^{80}$, the number of atoms in the universe

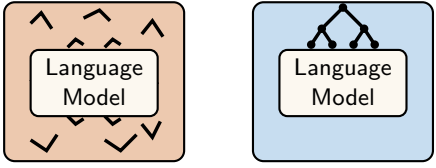
The Central Question

What are the abilities and limitations of LLMs?

The Sciences of the Artificial

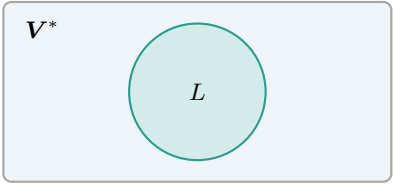
An artifact can be thought of as an interface between an 'inner' environment and an 'outer' environment. – Herbert Simon

An artifact can be thought of as an interface between an 'inner' environment and an 'outer' environment. – Herbert Simon



Formal Language

A formal language is a set (distribution) of strings generated by a formal grammar



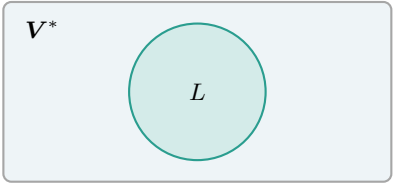
A formal language inside the universe of strings

Formal Language

A formal language is a set (distribution) of strings generated by a formal grammar

$S \to A18\ A16\ [0.50]$ } Level 4
$S \to A16\ A18\ A17\ [0.50]$ } Level 4
$A18 \to A15\ A14\ A13\ [0.50]$
$A18 \to A14\ A15\ A13\ [0.50]$
$A17 \to A14\ A13\ A15\ [0.50]$ } Level 3
$A17 \to A13\ A14\ A15\ [0.50]$ } Level 3
$A16 \to A14\ A15\ [0.50]$ } Level 3
$A16 \to A15\ A14\ [0.50]$ } Level 3
$A15 \to A11\ A12\ A10\ [0.50]$
$A15 \to A12\ A10\ A11\ [0.50]$
$A14 \to A11\ A10\ A12\ [0.50]$ } Level 2
$A14 \to A10\ A11\ A12\ [0.50]$ } Level 2
$A13 \to A10\ A12\ A11\ [0.50]$ } Level 2
$A13 \to A12\ A11\ A10\ [0.50]$ } Level 2
$A12 \to 9\ 8\ 7\ [0.50]$
$A12 \to 8\ 7\ [0.50]$
$A11 \to 6\ 5\ [0.50]$ } Level 1
$A11 \to 6\ 4\ 5\ [0.50]$ } Level 1
$A10 \to 3\ 1\ [0.50]$ } Level 1
$A10 \to 1\ 2\ 3\ [0.50]$ } Level 1

A hierarchical and probabilistic context-free grammar



A formal language inside the universe of strings



A sampled string with its derivation tree

Advantages of Formal Language

- **Precise specification.** We can decide whether a string belongs to the language
- **Exact sampling.** Train and test strings are sampled i.i.d.
- **No semantic ambiguity, no contamination.** Pure syntax, and unseen in pretraining
- **Control over language representation.** The grammar is a knob on language complexity

Our Pursuit, in Three Axes

1

No assumption about the computational model

Which languages are learnable at all?

2

A simplified model of the transformer

What can a transformer express and learn?

3

LLMs in practice

Can we explain the phenomena of LLMs related to language learning?

abstract

concrete

Axis 1: No Assumption about the Computational Model

From positive examples alone, can *any* learner learn a language?

- Can we identify a language from a list of languages?
- Can we generate new strings from the unknown language?

Axis 2: A Simplified Model of the Transformer

What is the *computational model* of a transformer?

- What languages can the transformer express?
- Is expressivity a good proxy for learnability?
- Can we interpret a transformer?

Axis 3: LLMs in Practice

Can we explain the *phenomena* of LLMs related to language learning?

- What languages are difficult for LLMs to learn in practice?
- Which learning modes are most effective?
- What is the implicit world model of an LLM when learning?
- What is the interplay between learning and memorization?

Logistics

Website	https://courses.mpi-sws.org/fllm-ws26
Class	Weekly, Thursday, 2–3:30 pm, Room 112 (MPI-SWS)
Office hours	Weekly, Friday, 2–3:30 pm, Room 112 (MPI-SWS)
Email	fllm26instructors@mpi-sws.org

Grading



- **Presentation.** 45–60 minutes on your paper, then Q&A
- **Report.** At most 2 pages, due before the next class; best three count
- **Participation.** Engagement in the Q&A

Passing

A combined score of **at least 50%**

What We Expect

A good presentation

- Motivate the question and formally state it
- Go deep on one proof or experiment; explain why you find it important
- Explain your takeaways from the paper

A good report on the presented paper

- Discuss at least one aspect of the paper that an AI reviewer, the presenter, or the original authors have missed
- Introduce insights to motivate future research

Class participation

Read the paper **before** the session and ask questions — the more difficult, the better

Syllabus

Date	Topic	Reading
22 Oct	Short intro and Q&A	—
29 Oct	No class	—
Axis 1 — No assumption about the computational model		
5 Nov	Generation in the Limit	Kleinberg & Mullainathan, *Language Generation in the Limit*, NeurIPS 2024
Axis 2 — A simplified model of the transformer		
12 Nov	Model of a Transformer	Weiss, Goldberg & Yahav, *Thinking Like Transformers*, ICML 2021
19 Nov	Predicting Length Generalization	Zhou et al., *What Algorithms Can Transformers Learn? A Study in Length Generalization*, ICLR 2024
26 Nov	Recovering the Learned Program	Friedman, Wettig & Chen, *Learning Transformer Programs*, NeurIPS 2023
Axis 3 — LLMs in practice		
3 Dec	Difficulty of Learning	Liu et al., *Exposing Attention Glitches with Flip-Flop Language Modeling*, NeurIPS 2023
14 Jan	Implicit World Models	Vafa et al., *Evaluating the World Model Implicit in a Generative Model*, NeurIPS 2024
21 Jan	Learning Modes	Ghosh et al., *Fine-Tuning vs. In-Context Learning in Large Language Models: A Formal Language Learning Perspective*, ACL 2026
28 Jan	Memorization & Learning	Ghosh et al., *Understanding the Interplay between Memorization and Learning in Large Language Models*, NeurIPS 2026

Next Steps



Before you leave today

Select at least three papers you would be happy to present, by 23 October

We assign papers on 26 October

https://courses.mpi-sws.org/fllm-ws26/

Background Reading

Optional, if either half of the seminar is new to you

Formal languages

Kim, *The Relationship Between the Chomsky Hierarchy and Automata*, 2019

Finite automata, pushdown automata, Turing machines, and the language classes they recognise

math.uchicago.edu/~may/REU2019/REUPapers/Kim.pdf

Large language models

Alammar, *The Illustrated Transformer*, 2018

Attention and the transformer architecture, explained visually

jalammar.github.io/illustrated-transformer